

Qw2

1. (a)

$$m = 64 \text{ bits},$$

$$n = 64 \text{ bits},$$

$$p = 64 \text{ bits} \quad (\text{Not sure})$$

$$k = 2^6 - 1 = 64 - 1 = 63$$

$$i = 2^7 - 1 = 128 - 1 = 127$$

$$j = 2^7 - 1 = 128 - 1 = 127$$

(Ans)

(b)

Virtual Address:  $0 \times E79EAA5B$

$$\begin{array}{ccccccc} 1110 & 0111 & 1001 & 1110 & 1010 & 1010 & 0101 & 1011 \\ \hline 57 & 121 & 106 & & & & 2643 & \end{array}$$

Page table Entry Size: 4 byte

$$\text{page size: } 2^{12} = 4096 \text{ bytes (4KB)}$$

First-Level page: First-level Entry Address =  $0 \times 59382453 + (57 \times 4)_d$   
 $= 0 \times 59382537$

Second-Level page: Second-level Entry Address =  $0 \times 7A432000 + (121 \times 4)_d$   
 $= 0 \times 7A4321E4$

Third-Level page: Third-level Entry Address =  $0 \times 8C120000 + (106 \times 4)_d$   
 $= 0 \times 8C1201A8$

$\therefore$  Final physical address =  $0 \times 9A560000 + 0 \times A53 \rightarrow (2643 \times 4)_d$   
 $= 0 \times 9A560A53$

(Ans)

02. `#!/usr/bin/bash`

`if [ "$#" -ne 2 ]; then`

`echo " usage: $0 <old-extension> <new-extension>"`

`exit 1`

`fi`

`old_ext = "$1"`

`new_ext = "$2"`

`for file in $(ls "$old_ext"); do`

`# check if any files match the pattern`

`if [ -e "$file" ]; then`

`base_name="${file%$old_ext}" # extract the filename without extension`

`mv "$file" "${base_name}$new_ext"`

`fi`

`done`

`echo " All files with extension $old_ext have been renamed to $new_ext."`

03. (a) `#!/bin/bash`

`file="filename.txt"`

`if [ -f "$file" ]; then`

`line_count=$(wc -l < "$file")`

`if [ "$line_count" -gt 100 ]; then`

`sed -n '15,22p' "$file"`

`else`

`echo "The file '$file' has fewer than 100 lines."`

`fi`

`fi`

(b)

```
if command -v history > /dev/null 2>&1; then ⇒ if [ $(history | grep -c 'grep') -gt 0 ]; then
    history | grep 'grep' > grep.txt
else
    echo "The 'historygrep' not found in history"
fi
```

09.

(a) (i) No Step-level Control: The model doesn't properly manage individual steps — it only controls the total number of people moving upward or downward.

(ii) Overtaking and Safety condition: there is no mechanism to prevent overtaking, or to ensure that a person can move to the next step only if the step is empty.

(iii) Potential for Deadlock: Since there is no restriction on the sequence of moving, deadlock can occur.

(b) Three types of semaphore:

(i) Step control semaphore: Type: counting semaphore, initial value: 2 (each step can have at most 2 people — upward / downward)

(ii) Movement control semaphore: counting semaphore, initial value:  $\infty$  (to control total number of people moving in each direction)

(iii) Binary semaphore for critical section: in Mutex, initial value: 1, to prevent race conditions when checking and updating the status of the current or next step.

05.

किसे पूछा है ? 

उत्तर last assignment की हिन, कर्म गरीब ?



06.

It is not a good idea to omit the `sleep()`.

(i) Interrupts are necessary for scheduling; for preemptive multitasking.

(ii) If interrupts are disabled, clock ticks will not occur

and process preemption will not happen as expected,

(iii) The kernel might not handle important asynchronous events like I/O interrupt.

07. modified version of `wait()` system call can act incorrectly due to the ordering of conditions.

If a child process completes execution and changes its state to exited, then the parent first checks if there is no children or if it has been killed, finds neither of condition is true.

Then the parent calls `sleep()` to wait for a child process to exit. During the time, the parent goes to ~~sleep~~ sleep, it misses the fact that a child process has already exited.

the issues are:

1. Race condition Leading to Missed Exited child.
2. Incorrect Return value
3. No proper Synchronization

MCQ.

1. (d)
2. (b) (c)
3. (c) pthread\_join
4. (b) writers
5. (b) g
6. (c) Kernel bug
7. (c)
8. (d)